# LAB SHEET 3 — MONGODB APPLICATIONS

Only Lab Sheet 3 code from the supplied NoSQL PDF. Run in mongosh.

## Database setup

```
use Lab3DB
```

## Q1. Student Activity Tracking System.

```
db.students.insertMany([
{sid:1,name:'Aman',courses:['DBMS','Python'],attendance:85,skills:['SQL','Python'],activities:[{type:'quiz',score:90},{ty
{sid:2,name:'Riya',courses:['DBMS','Java'],attendance:92,skills:['Java','SQL'],activities:[{type:'quiz',score:95}]},
{sid:3,name:'Karan',courses:['Python'],attendance:65,skills:['Python'],activities:[{type:'lab',score:70}]}
])
db.students.find({attendance:{$gte:75}})
db.students.find({courses:'DBMS'})
db.students.find({'activities.type':'quiz','activities.score':{$gte:90}})
db.students.find({skills:'Python'})
```

## Q2. Event Logging System.

```
db.events.insertMany([
{user:'aman',event:'login',time:new Date(),status:'success'},
{user:'riya',event:'login',time:new Date(),status:'failed'},
{user:'riya',event:'login',time:new Date(),status:'failed'},
{user:'aman',event:'file-upload',time:new Date(),status:'success'},
{user:'riya',event:'password-change',time:new Date(),status:'success'}
])
db.events.aggregate([
{$match:{event:'login',status:'failed'}},
{$group:{_id:'$user',failedAttempts:{$sum:1}}},
{$match:{failedAttempts:{$gte:2}}}
])
db.events.aggregate([{$group:{_id:'$event',count:{$sum:1}}},{$sort:{count:-1}}])
```

## Q3. Mini Blogging Platform.

```
db.blogs.insertMany([
{title:'Mongo Basics',author:'Aman',category:'Technology',tags:['mongo','database'],comments:[{by:'Riya',text:'Good'}],pu
{title:'Java Intro',author:'Riya',category:'Programming',tags:['java','coding'],comments:[],published:true,date:new Date(
{title:'NoSQL Notes',author:'Aman',category:'Technology',tags:['nosql','database'],comments:[],published:false,date:new D
])
db.blogs.find({tags:'database'})
db.blogs.find({category:'Technology',published:true})
db.blogs.find({tags:{$all:['mongo','database']}})
db.blogs.find({'comments.by':'Riya'})
```

## Q4. Flexible Product Catalog.

```
db.products.insertMany([
{name:'Laptop A',category:'laptop',brand:'Dell',ram:16,storage:512,price:55000},
{name:'Phone B',category:'mobile',brand:'Samsung',camera:50,battery:5000,price:22000},
{name:'Camera C',category:'camera',brand:'Canon',megapixels:24,lens:'18-55mm',price:40000}
])
db.products.find({category:'laptop',ram:{$gte:8}})
db.products.find({category:'mobile',battery:{$gte:4500}})
db.products.find({category:'camera',megapixels:{$gte:20}})
```

## Q5. E-Commerce Order Management System.

```
db.orders.insertMany([
{orderId:1,customerId:101,customer:'Aman',items:[{product:'Mouse',qty:2,price:400},{product:'USB',qty:1,price:250}],statu
{orderId:2,customerId:102,customer:'Riya',items:[{product:'Laptop',qty:1,price:45000}],status:'shipped'}
])
db.orders.find({customerId:101})
db.orders.aggregate([
{$unwind:'$items'},
{$group:{_id:{customerId:'$customerId',customer:'$customer'},total:{$sum:{$multiply:['$items.qty','$items.price']}}}}
])
```

## Q6. Website Analytics System.

```
db.visits.insertMany([
{page:'/home',user:1,device:'mobile',browser:'Chrome',location:'India',time:new Date()},
```

```
{page:'/about',user:2,device:'desktop',browser:'Edge',location:'India',time:new Date()},
{page:'/home',user:3,device:'mobile',browser:'Chrome',location:'India',time:new Date()},
{page:'/products',user:1,device:'mobile',browser:'Chrome',location:'India',time:new Date()}
])
db.visits.aggregate([{$group:{_id:'$page',views:{$sum:1}}},{$sort:{views:-1}}])
db.visits.aggregate([{$group:{_id:'$device',visits:{$sum:1}}},{$sort:{visits:-1}}])
```

## Q7. Content Management System.

```
db.articles.insertMany([
{title:'DBMS',author:'Aman',category:'Study',tags:['dbms'],published:true},
{title:'MongoDB',author:'Riya',category:'Technology',tags:['mongo','nosql'],published:false},
{title:'Python',author:'Aman',category:'Study',tags:['python'],published:true}
])
db.articles.find({published:true})
db.articles.find({published:false})
db.articles.find({category:'Study',tags:'python'})
db.articles.updateOne({title:'MongoDB'},{$set:{published:true}})
```

## Q8. Shopping Cart using embedded documents.

```
db.carts.insertOne({userId:101,items:[{productId:1,name:'Mouse',qty:2,price:400},{productId:2,name:'USB',qty:1,price:250}
db.carts.updateOne({userId:101},{$push:{items:{productId:3,name:'Keyboard',qty:1,price:800}}})
db.carts.updateOne({userId:101,'items.productId':1},{$set:{'items.$.qty':3}})
db.carts.updateOne({userId:101},{$pull:{items:{productId:2}}})
db.carts.aggregate([{$match:{userId:101}},{$unwind:'$items'},{$group:{_id:'$userId',cartTotal:{$sum:{$multiply:['$items.c
```

## Q9. Product Review and Rating System.

```
db.reviews.insertMany([
{productId:1,user:'Aman',rating:5,comment:'Excellent'},
{productId:1,user:'Riya',rating:4,comment:'Good'},
{productId:2,user:'Karan',rating:3,comment:'Average'}
])
db.reviews.aggregate([
{$group:{_id:'$productId',averageRating:{$avg:'$rating'},reviewCount:{$sum:1}}}
])
```

## Q10. Inventory Monitoring System.

```
db.inventory.insertMany([
{product:'Mouse',stock:5,minStock:10,supplier:'ABC',restockDate:new Date()},
{product:'Keyboard',stock:30,minStock:10,supplier:'XYZ',restockDate:new Date()},
{product:'Monitor',stock:3,minStock:5,supplier:'ABC',restockDate:new Date()}
])
db.inventory.find({$expr:{$lte:['$stock','$minStock']}})
db.inventory.updateOne({product:'Mouse'},{$inc:{stock:20}})
```

## Q11. Bank Wallet Transfer using MongoDB transaction.

```
// Example in mongosh connected to a replica set
const session = db.getMongo().startSession();
const bank = session.getDatabase('Lab3DB');
session.startTransaction();
try {
bank.wallets.updateOne({user:'Aman',balance:{$gte:500}},{$inc:{balance:-500}});
bank.wallets.updateOne({user:'Riya'},{$inc:{balance:500}});
session.commitTransaction();
} catch (e) {
session.abortTransaction();
throw e;
} finally {
session.endSession();
}
// Create sample wallets first:
db.wallets.insertMany([{user:'Aman',balance:2000},{user:'Riya',balance:1000}])
```

## Q12. User Notification System.

```
db.notifications.insertMany([
{userId:1,message:'Order placed',type:'order',read:false,createdAt:new Date()},
{userId:1,message:'Welcome',type:'system',read:true,createdAt:new Date()},
{userId:2,message:'New message',type:'message',read:false,createdAt:new Date()}
])
db.notifications.find({userId:1,read:false})
db.notifications.updateMany({userId:1,read:false},{$set:{read:true}})
db.notifications.find({userId:1})
```

## Q13. Course Learning Platform.

```
db.learning.insertMany([
{student:'Aman',course:'MongoDB',instructor:'Mr X',modules:[{name:'Intro',done:true},{name:'CRUD',done:true}],progress:10
{student:'Riya',course:'MongoDB',instructor:'Mr X',modules:[{name:'Intro',done:true},{name:'CRUD',done:false}],progress:5
{student:'Karan',course:'Python',instructor:'Ms Y',modules:[{name:'Basics',done:true}],progress:80,completed:false}
])
db.learning.find({course:'MongoDB',progress:{$gte:50}})
db.learning.find({completed:true})
db.learning.find({progress:{$lt:100}})
```

## Q14. Real-Time Order Tracking System.

```
db.tracking.insertOne({orderId:501,status:'Placed',history:[{status:'Placed',at:new Date()}]})
db.tracking.updateOne({orderId:501},{$set:{status:'Confirmed'},$push:{history:{status:'Confirmed',at:new Date()}}})
db.tracking.updateOne({orderId:501},{$set:{status:'Shipped'},$push:{history:{status:'Shipped',at:new Date()}}})
db.tracking.updateOne({orderId:501},{$set:{status:'Delivered'},$push:{history:{status:'Delivered',at:new Date()}}})
db.tracking.findOne({orderId:501})
```

## Q15. Flexible News Database.

```
db.news.insertMany([
{title:'Tech Update',author:'Aman',location:'India',category:'Technology',tags:['AI','tech'],image:'ai.jpg',source:'News
{title:'City News',location:'Roorkee',category:'Local',tags:['city'],source:'News B'},
{title:'Sports Update',author:'Riya',category:'Sports',tags:['cricket'],source:'News C'}
])
db.news.find({category:'Technology'})
db.news.find({location:'Roorkee'})
db.news.find({tags:'cricket'})
db.news.find({author:{$exists:true}})
```